Concurrent Multi-Device Attack Simulations

Project: Agentic AI Cyberdefence
Date: 7 October 2026

Scenario 1: Three Attacks with No Human Input

Input: Raw Attack Logs

```
raw_logs_campaign = [
    # Stage 1: Compromised Office Workstation in Administration Zone
    {
        "timestamp": "2026-10-07T10:00:00Z",
        "source": "admin_pcap_sensor",
        "event_type": "SMB_FLOW",
        "src": "10.0.3.15",  # Attacker IP
        "dst": "10.0.3.20",
        "protocol": "TCP",
        "bytes": 80,
        "zone": "administration",
        "scenario_id": "severe_ransomware_campaign"
    },
    # Stage 2: Lateral Spread to Laboratory Server
    {
        "timestamp": "2026-10-07T10:02:00Z",
        "source": "lab_pcap_sensor",
        "event_type": "PORT_SCAN",
        "src": "10.0.3.15",  # Same Attacker IP
        "dst": "10.0.2.10",
        "protocol": "TCP",
        "bytes": 1200,
        "zone": "laboratory",
        "scenario_id": "severe_ransomware_campaign"
    },
    # Stage 3: Exploit Attempt targeting Life-Critical ICU Patient Monitor Gateway
    {
        "timestamp": "2026-10-07T10:05:00Z",
        "source": "icu_pcap_sensor",
        "event_type": "EXPLOIT_ATTEMPT",
        "src": "10.0.3.15",  # Same Attacker IP targeting Tier 1!
        "dst": "10.0.1.45",
        "protocol": "TCP",
        "bytes": 4096,
        "zone": "intensive_care",
        "scenario_id": "severe_ransomware_campaign"
    }
]
```

Output: Simulation Run

```
(agentic-ai-cyberdefence) soumadipsen@pop-os:~/Desktop/AI Engineering/agentic-ai-
cyberdefence$ uv run python main.py
================================================================================
 ACCDS END-TO-END PIPELINE: MULTI-STAGE LATERAL CAMPAIGN (LAYERS 0 - 6)
================================================================================


================================================================================
 >>> PROCESSING CAMPAIGN STAGE 1: Raw Log from Zone 'administration'
================================================================================
--- [1] LAYER 0: NORMALIZED OBSERVATION EVENT ---
{
  "event_id": "EVT-9D46F7B7",
  "timestamp": "2026-10-07T10:00:00+00:00",
  "source": "admin_pcap_sensor",
  "event_type": "SMB_FLOW",
  "subject": "10.0.3.15",
  "src": "10.0.3.15",
  "dst": "10.0.3.20",
  "protocol": "TCP",
  "bytes": 80,
  "zone": "administration",
  "scenario_id": "aiims_lateral_campaign",
  "raw_reference": "/logs/raw/aiims_lateral_campaign/EVT-9D46F7B7.json"
}

--- [2] LAYER 1: ASSET CONTEXT & CLINICAL TIER ---
{
  "asset_id": "DEV-OFFICE-LAPTOP-04",
  "asset_type": "Reception Desk Endpoint",
  "owner_zone": "Administration",
  "vendor": "Dell",
  "software_version": "Windows 11",
  "clinical_tier": 3,
  "known_constraints": [
    "Non-clinical endpoint"
  ],
  "allowed_peers": [
    "10.0.3.1"
  ],
  "normal_services": [
    "HTTPS",
    "SMB"
  ],
  "fallback_mode": "NONE",
  "identity_confidence": 0.99,
  "timestamp": "2026-10-07T10:00:00+00:00"
}

--- [3] LAYER 2: DETECTION FINDING ---
{
  "finding_id": "FIND-001",
```

```
  "asset_id": "DEV-OFFICE-LAPTOP-04",
  "attacker_src": "10.0.3.15",
  "behavior_window": {
    "start": "2026-10-07T10:00:00+00:00",
    "end": "2026-10-07T10:05:00+00:00"
  },
  "anomaly_score": 0.65,
  "risk_score": 0.65,
  "indicators": [
    "UNEXPECTED_SMB_TRAFFIC"
  ],
  "baseline_version": "v1.0.0",
  "confidence": 0.95,
  "related_events": [
    "EVT-9D46F7B7"
  ],
  "recommended_observation_period": 60,
  "timestamp": "2026-10-07T10:00:00+00:00"
}

--- [4] LAYER 3: CORRELATED CAMPAIGN RESPONSE PROPOSAL ---
Case ID: CASE-CORR-8B9AAF
Affected Assets in Campaign: ['DEV-OFFICE-LAPTOP-04']
Approval Level: AUTO_APPROVED
{
  "case_id": "CASE-CORR-8B9AAF",
  "threat_summary": "Detected anomalous behavior (UNEXPECTED_SMB_TRAFFIC) targeting asset
DEV-OFFICE-LAPTOP-04.",
  "affected_assets": [
    "DEV-OFFICE-LAPTOP-04"
  ],
  "candidate_actions": [
    {
      "action_type": "COARSE_ISOLATION",
      "target_asset": "DEV-OFFICE-LAPTOP-04",
      "blocked_flows": [],
      "preserved_flows": []
    }
  ],
  "clinical_impact": "LOW: All affected assets are non-clinical Tier 3. Rapid isolation
allowed.",
  "confidence": 0.95,
  "reason": "Case CASE-CORR-8B9AAF contains 1 correlated finding(s). Max Risk Score:
0.65. Safety mode: NORMAL.",
  "approval_level": "AUTO_APPROVED",
  "expiry_time": "2026-10-07T13:35:42.031724+00:00",
  "rollback_plan": {
    "rollback_token_spec": "RB-CAMPAIGN-B04620",
    "steps": [
      "Reconnect isolated endpoint 'DEV-OFFICE-LAPTOP-04'",
      "Verify telemetry heartbeat across all restored hosts"
    ],
    "auto_rollback_seconds": 300
  },
```

```
  "correlated_finding_ids": [
    "FIND-001"
  ]
}

================================================================================
 >>> PROCESSING CAMPAIGN STAGE 2: Raw Log from Zone 'laboratory'
================================================================================
--- [1] LAYER 0: NORMALIZED OBSERVATION EVENT ---
{
  "event_id": "EVT-787E5B3F",
  "timestamp": "2026-10-07T10:02:00+00:00",
  "source": "lab_pcap_sensor",
  "event_type": "PORT_SCAN",
  "subject": "10.0.3.15",
  "src": "10.0.3.15",
  "dst": "10.0.2.10",
  "protocol": "TCP",
  "bytes": 1200,
  "zone": "laboratory",
  "scenario_id": "aiims_lateral_campaign",
  "raw_reference": "/logs/raw/aiims_lateral_campaign/EVT-787E5B3F.json"
}

--- [2] LAYER 1: ASSET CONTEXT & CLINICAL TIER ---
{
  "asset_id": "DEV-OFFICE-LAPTOP-04",
  "asset_type": "Reception Desk Endpoint",
  "owner_zone": "Administration",
  "vendor": "Dell",
  "software_version": "Windows 11",
  "clinical_tier": 3,
  "known_constraints": [
    "Non-clinical endpoint"
  ],
  "allowed_peers": [
    "10.0.3.1"
  ],
  "normal_services": [
    "HTTPS",
    "SMB"
  ],
  "fallback_mode": "NONE",
  "identity_confidence": 0.99,
  "timestamp": "2026-10-07T10:02:00+00:00"
}

--- [3] LAYER 2: DETECTION FINDING ---
{
  "finding_id": "FIND-002",
  "asset_id": "DEV-OFFICE-LAPTOP-04",
  "attacker_src": "10.0.3.15",
  "behavior_window": {
    "start": "2026-10-07T10:00:00+00:00",
```

```
    "end": "2026-10-07T10:05:00+00:00"
  },
  "anomaly_score": 0.82,
  "risk_score": 0.82,
  "indicators": [
    "PORT_SCAN_SWEEP",
    "UNAUTHORIZED_PEER_CONNECT"
  ],
  "baseline_version": "v1.0.0",
  "confidence": 0.95,
  "related_events": [
    "EVT-787E5B3F"
  ],
  "recommended_observation_period": 60,
  "timestamp": "2026-10-07T10:02:00+00:00"
}

--- [4] LAYER 3: CORRELATED CAMPAIGN RESPONSE PROPOSAL ---
Case ID: CASE-CORR-8B9AAF
Affected Assets in Campaign: ['DEV-OFFICE-LAPTOP-04']
Approval Level: AUTO_APPROVED
{
  "case_id": "CASE-CORR-8B9AAF",
  "threat_summary": "LATERAL CAMPAIGN DETECTED: Correlated 2 anomalies across assets
[DEV-OFFICE-LAPTOP-04]. Attacker origin(s): [10.0.3.15]. Potential multi-stage attack
chain in progress.",
  "affected_assets": [
    "DEV-OFFICE-LAPTOP-04"
  ],
  "candidate_actions": [
    {
      "action_type": "COARSE_ISOLATION",
      "target_asset": "DEV-OFFICE-LAPTOP-04",
      "blocked_flows": [],
      "preserved_flows": []
    }
  ],
  "clinical_impact": "LOW: All affected assets are non-clinical Tier 3. Rapid isolation
allowed.",
  "confidence": 0.95,
  "reason": "Case CASE-CORR-8B9AAF contains 2 correlated finding(s). Max Risk Score:
0.82. Safety mode: NORMAL.",
  "approval_level": "AUTO_APPROVED",
  "expiry_time": "2026-10-07T13:35:42.032360+00:00",
  "rollback_plan": {
    "rollback_token_spec": "RB-CAMPAIGN-3B0098",
    "steps": [
      "Reconnect isolated endpoint 'DEV-OFFICE-LAPTOP-04'",
      "Verify telemetry heartbeat across all restored hosts"
    ],
    "auto_rollback_seconds": 300
  },
  "correlated_finding_ids": [
    "FIND-001",
```

```
    "FIND-002"
  ]
}

================================================================================
 >>> PROCESSING CAMPAIGN STAGE 3: Raw Log from Zone 'intensive_care'
================================================================================
--- [1] LAYER 0: NORMALIZED OBSERVATION EVENT ---
{
  "event_id": "EVT-E48F3576",
  "timestamp": "2026-10-07T10:05:00+00:00",
  "source": "icu_pcap_sensor",
  "event_type": "EXPLOIT_ATTEMPT",
  "subject": "10.0.3.15",
  "src": "10.0.3.15",
  "dst": "10.0.1.45",
  "protocol": "TCP",
  "bytes": 4096,
  "zone": "intensive_care",
  "scenario_id": "aiims_lateral_campaign",
  "raw_reference": "/logs/raw/aiims_lateral_campaign/EVT-E48F3576.json"
}

--- [2] LAYER 1: ASSET CONTEXT & CLINICAL TIER ---
{
  "asset_id": "DEV-OFFICE-LAPTOP-04",
  "asset_type": "Reception Desk Endpoint",
  "owner_zone": "Administration",
  "vendor": "Dell",
  "software_version": "Windows 11",
  "clinical_tier": 3,
  "known_constraints": [
    "Non-clinical endpoint"
  ],
  "allowed_peers": [
    "10.0.3.1"
  ],
  "normal_services": [
    "HTTPS",
    "SMB"
  ],
  "fallback_mode": "NONE",
  "identity_confidence": 0.99,
  "timestamp": "2026-10-07T10:05:00+00:00"
}

--- [3] LAYER 2: DETECTION FINDING ---
{
  "finding_id": "FIND-003",
  "asset_id": "DEV-OFFICE-LAPTOP-04",
  "attacker_src": "10.0.3.15",
  "behavior_window": {
    "start": "2026-10-07T10:00:00+00:00",
    "end": "2026-10-07T10:05:00+00:00"
```

```
  },
  "anomaly_score": 0.95,
  "risk_score": 0.95,
  "indicators": [
    "EXPLOIT_ATTEMPT_SMB",
    "CRITICAL_FLOW_INTERRUPTION_RISK"
  ],
  "baseline_version": "v1.0.0",
  "confidence": 0.95,
  "related_events": [
    "EVT-E48F3576"
  ],
  "recommended_observation_period": 60,
  "timestamp": "2026-10-07T10:05:00+00:00"
}

--- [4] LAYER 3: CORRELATED CAMPAIGN RESPONSE PROPOSAL ---
Case ID: CASE-CORR-8B9AAF
Affected Assets in Campaign: ['DEV-OFFICE-LAPTOP-04']
Approval Level: AUTO_APPROVED
{
  "case_id": "CASE-CORR-8B9AAF",
  "threat_summary": "LATERAL CAMPAIGN DETECTED: Correlated 3 anomalies across assets
[DEV-OFFICE-LAPTOP-04]. Attacker origin(s): [10.0.3.15]. Potential multi-stage attack
chain in progress.",
  "affected_assets": [
    "DEV-OFFICE-LAPTOP-04"
  ],
  "candidate_actions": [
    {
      "action_type": "COARSE_ISOLATION",
      "target_asset": "DEV-OFFICE-LAPTOP-04",
      "blocked_flows": [],
      "preserved_flows": []
    }
  ],
  "clinical_impact": "LOW: All affected assets are non-clinical Tier 3. Rapid isolation
allowed.",
  "confidence": 0.95,
  "reason": "Case CASE-CORR-8B9AAF contains 3 correlated finding(s). Max Risk Score:
0.95. Safety mode: NORMAL.",
  "approval_level": "AUTO_APPROVED",
  "expiry_time": "2026-10-07T13:35:42.032933+00:00",
  "rollback_plan": {
    "rollback_token_spec": "RB-CAMPAIGN-461273",
    "steps": [
      "Reconnect isolated endpoint 'DEV-OFFICE-LAPTOP-04'",
      "Verify telemetry heartbeat across all restored hosts"
    ],
    "auto_rollback_seconds": 300
  },
  "correlated_finding_ids": [
    "FIND-001",
    "FIND-002",
```

```
    "FIND-003"
  ]
}

================================================================================
--- [6] LAYER 4: ENFORCEMENT ENGINE EXECUTION ---
================================================================================
{
  "action_id": "ACT-CAMPAIGN-EXEC-01",
  "case_id": "CASE-CORR-8B9AAF",
  "decision_id": "CASE-CORR-8B9AAF",
  "target_asset": "DEV-OFFICE-LAPTOP-04",
  "status": "APPLIED",
  "applied_controls": [
    "COARSE_ISOLATION",
    "SURGICAL_BLOCK"
  ],
  "blocked_flows": [
    {
      "protocol": "TCP",
      "port": 445
    }
  ],
  "preserved_flows": [
    {
      "protocol": "HL7",
      "port": 2575
    }
  ],
  "start_time": "2026-10-07T10:06:05.000000+00:00",
  "expiry_time": "2026-10-07T13:35:42.032933+00:00",
  "verification": {
    "malicious_flow_blocked": true,
    "preserved_clinical_flows_available": true,
    "checked_at": "2026-10-07T10:06:10.000000+00:00"
  },
  "rollback_token": "RB-CAMPAIGN-461273"
}

================================================================================
--- [7] LAYER 6: COMPLIANCE, REDACTION & EVIDENCE EXPORT ---
================================================================================
📄 Incident Timeline Report written to: ['artifacts/full_run/RPT-CASE-CORR-8B9AAF.md',
'artifacts/full_run/RPT-CASE-CORR-8B9AAF.json']
📦 Evidence Package exported with ID: EVID-CASE-CORR-8B9AAF
🔒 SHA-256 Tamper-Evident Audit Chain verified and saved to disk.
(agentic-ai-cyberdefence) soumadipsen@pop-os:~/Desktop/AI Engineering/agentic-ai-
cyberdefence$
```

Scenario 2: Six Attacks with Human Validation

for a Tier 1 Device

Input: Raw Attack Logs

```
raw_logs_campaign = [
    # Stage 1: SMB Discovery/Reconnaissance on Admin Workstation
    {
        "timestamp": "2026-10-07T10:00:00Z",
        "source": "admin_pcap_sensor",
        "event_type": "SMB_FLOW",
        "src": "10.0.3.15",  # Primary Attacker Source
        "dst": "10.0.3.20",
        "protocol": "TCP",
        "bytes": 80,
        "zone": "administration",
        "scenario_id": "six_stage_attack_campaign"
    },
    # Stage 2: TCP Port Scan Sweep targeting Radiology Workstation
    {
        "timestamp": "2026-10-07T10:02:00Z",
        "source": "radiology_pcap_sensor",
        "event_type": "PORT_SCAN",
        "src": "10.0.3.15",  # Same Attacker Source
        "dst": "10.0.4.12",
        "protocol": "TCP",
        "bytes": 1200,
        "zone": "radiology",
        "scenario_id": "six_stage_attack_campaign"
    },
    # Stage 3: Credential Dumping Attempt on PACS Server (Shares EVT-SHARED-99 with Stage
2)
    {
        "timestamp": "2026-10-07T10:03:00Z",
        "source": "pacs_auth_sensor",
        "event_type": "CREDENTIAL_DUMP",
        "src": "10.0.4.88",
        "dst": "10.0.4.10",
        "protocol": "TCP",
        "bytes": 3400,
        "zone": "radiology",
        "scenario_id": "six_stage_attack_campaign"
    },
    # Stage 4: Unauthorized DB Query / Malware Stage on Lab Server
    {
        "timestamp": "2026-10-07T10:05:00Z",
        "source": "lab_pcap_sensor",
        "event_type": "MALWARE_STAGING",
        "src": "10.0.3.15",  # Same Attacker Source
        "dst": "10.0.2.10",  # lab-server compromised here
        "protocol": "TCP",
        "bytes": 8192,
        "zone": "laboratory",
```

```
        "scenario_id": "six_stage_attack_campaign"
    },
    # Stage 5: Lateral Exploitation Attempt on Patient Monitor Gateway (Chain from lab-
server!)
    {
        "timestamp": "2026-10-07T10:07:00Z",
        "source": "icu_pcap_sensor",
        "event_type": "EXPLOIT_ATTEMPT",
        "src": "lab-server",  # LATERAL CHAIN: Compromised lab-server attacks ICU!
        "dst": "10.0.1.45",   # icu-monitor-gw (Tier 1)
        "protocol": "TCP",
        "bytes": 4096,
        "zone": "intensive_care",
        "scenario_id": "six_stage_attack_campaign"
    },
    # Stage 6: Command & Control (C2) / HL7 Disruption on ICU Controller (Pivot from ICU
Monitor)
    {
        "timestamp": "2026-10-07T10:09:00Z",
        "source": "icu_pcap_sensor",
        "event_type": "C2_BEACON",
        "src": "icu-monitor-gw",  # LATERAL CHAIN: Pivot from icu-monitor-gw to icu-
controller!
        "dst": "10.0.1.50",       # icu-controller (Tier 1)
        "protocol": "TCP",
        "bytes": 1024,
        "zone": "intensive_care",
        "scenario_id": "six_stage_attack_campaign"
    }
]
```

Output 1: All Computation Before Validation

```
(agentic-ai-cyberdefence) soumadipsen@pop-os:~/Desktop/AI Engineering/agentic-ai-
cyberdefence$ uv run python main.py
================================================================================
 ACCDS SEVERE ATTACK SIMULATION: 6-STAGE LATERAL CAMPAIGN (LAYERS 0 - 6)
================================================================================


================================================================================
 >>> PROCESSING CAMPAIGN STAGE 1/6: Event Type 'SMB_FLOW' in Zone 'administration'
================================================================================

--- [1] LAYER 0: NORMALIZED OBSERVATION EVENT ---
{
  "event_id": "EVT-6A0D639A",
  "timestamp": "2026-10-07T10:00:00+00:00",
  "source": "admin_pcap_sensor",
  "event_type": "SMB_FLOW",
  "subject": "10.0.3.15",
  "src": "10.0.3.15",
  "dst": "10.0.3.20",
  "protocol": "TCP",
  "bytes": 80,
  "zone": "administration",
  "scenario_id": "six_stage_attack_campaign",
  "raw_reference": "/logs/raw/six_stage_attack_campaign/EVT-6A0D639A.json"
}

--- [2] LAYER 1: ASSET CONTEXT & CLINICAL TIER ---
{
  "asset_id": "DEV-OFFICE-LAPTOP-04",
  "asset_type": "Reception Desk Endpoint",
  "owner_zone": "Administration",
  "vendor": "Dell",
  "software_version": "Windows 11",
  "clinical_tier": 3,
  "known_constraints": [
    "Non-clinical endpoint"
  ],
  "allowed_peers": [
    "10.0.3.1"
  ],
  "normal_services": [
    "HTTPS",
    "SMB"
  ],
  "fallback_mode": "NONE",
  "identity_confidence": 0.99,
  "timestamp": "2026-10-07T10:00:00+00:00"
}

--- [3] LAYER 2: DETECTION FINDING ---
{
```

```
  "finding_id": "FIND-001",
  "asset_id": "DEV-OFFICE-LAPTOP-04",
  "attacker_src": "10.0.3.15",
  "behavior_window": {
    "start": "2026-10-07T10:00:00+00:00",
    "end": "2026-10-07T10:09:00+00:00"
  },
  "anomaly_score": 0.6,
  "risk_score": 0.6,
  "indicators": [
    "UNEXPECTED_SMB_TRAFFIC"
  ],
  "baseline_version": "v1.0.0",
  "confidence": 0.95,
  "related_events": [
    "EVT-6A0D639A"
  ],
  "recommended_observation_period": 60,
  "timestamp": "2026-10-07T10:00:00+00:00"
}

--- [4] LAYER 3: CORRELATED CAMPAIGN RESPONSE PROPOSAL ---
Case ID:                    CASE-CORR-A6E68F
Correlated Affected Assets:  ['DEV-OFFICE-LAPTOP-04']
Correlated Finding Count:   1
Approval Level Required:     AUTO_APPROVED
{
  "case_id": "CASE-CORR-A6E68F",
  "threat_summary": "Detected anomalous behavior (UNEXPECTED_SMB_TRAFFIC) targeting asset
DEV-OFFICE-LAPTOP-04.",
  "affected_assets": [
    "DEV-OFFICE-LAPTOP-04"
  ],
  "candidate_actions": [
    {
      "action_type": "COARSE_ISOLATION",
      "target_asset": "DEV-OFFICE-LAPTOP-04",
      "blocked_flows": [],
      "preserved_flows": []
    }
  ],
  "clinical_impact": "LOW: All affected assets are non-clinical Tier 3. Rapid isolation
allowed.",
  "confidence": 0.95,
  "reason": "Case CASE-CORR-A6E68F contains 1 correlated finding(s). Max Risk Score: 0.6.
Safety mode: NORMAL.",
  "approval_level": "AUTO_APPROVED",
  "expiry_time": "2026-10-07T14:03:48.231828+00:00",
  "rollback_plan": {
    "rollback_token_spec": "RB-CAMPAIGN-FFB8F8",
    "steps": [
      "Reconnect isolated endpoint 'DEV-OFFICE-LAPTOP-04'",
      "Verify telemetry heartbeat across all restored hosts"
    ],
```

```
    "auto_rollback_seconds": 300
  },
  "correlated_finding_ids": [
    "FIND-001"
  ]
}

================================================================================
 >>> PROCESSING CAMPAIGN STAGE 2/6: Event Type 'PORT_SCAN' in Zone 'radiology'
================================================================================

--- [1] LAYER 0: NORMALIZED OBSERVATION EVENT ---
{
  "event_id": "EVT-70F77E30",
  "timestamp": "2026-10-07T10:02:00+00:00",
  "source": "radiology_pcap_sensor",
  "event_type": "PORT_SCAN",
  "subject": "10.0.3.15",
  "src": "10.0.3.15",
  "dst": "10.0.4.12",
  "protocol": "TCP",
  "bytes": 1200,
  "zone": "radiology",
  "scenario_id": "six_stage_attack_campaign",
  "raw_reference": "/logs/raw/six_stage_attack_campaign/EVT-70F77E30.json"
}

--- [2] LAYER 1: ASSET CONTEXT & CLINICAL TIER ---
{
  "asset_id": "DEV-OFFICE-LAPTOP-04",
  "asset_type": "Reception Desk Endpoint",
  "owner_zone": "Administration",
  "vendor": "Dell",
  "software_version": "Windows 11",
  "clinical_tier": 3,
  "known_constraints": [
    "Non-clinical endpoint"
  ],
  "allowed_peers": [
    "10.0.3.1"
  ],
  "normal_services": [
    "HTTPS",
    "SMB"
  ],
  "fallback_mode": "NONE",
  "identity_confidence": 0.99,
  "timestamp": "2026-10-07T10:02:00+00:00"
}

--- [3] LAYER 2: DETECTION FINDING ---
{
  "finding_id": "FIND-002",
  "asset_id": "DEV-OFFICE-LAPTOP-04",
```

```
  "attacker_src": "10.0.3.15",
  "behavior_window": {
    "start": "2026-10-07T10:00:00+00:00",
    "end": "2026-10-07T10:09:00+00:00"
  },
  "anomaly_score": 0.72,
  "risk_score": 0.72,
  "indicators": [
    "PORT_SCAN_SWEEP"
  ],
  "baseline_version": "v1.0.0",
  "confidence": 0.95,
  "related_events": [
    "EVT-70F77E30",
    "EVT-SHARED-99"
  ],
  "recommended_observation_period": 60,
  "timestamp": "2026-10-07T10:02:00+00:00"
}

--- [4] LAYER 3: CORRELATED CAMPAIGN RESPONSE PROPOSAL ---
Case ID:                    CASE-CORR-A6E68F
Correlated Affected Assets:  ['DEV-OFFICE-LAPTOP-04']
Correlated Finding Count:   2
Approval Level Required:     AUTO_APPROVED
{
  "case_id": "CASE-CORR-A6E68F",
  "threat_summary": "LATERAL CAMPAIGN DETECTED: Correlated 2 anomalies across assets
[DEV-OFFICE-LAPTOP-04]. Attacker origin(s): [10.0.3.15]. Potential multi-stage attack
chain in progress.",
  "affected_assets": [
    "DEV-OFFICE-LAPTOP-04"
  ],
  "candidate_actions": [
    {
      "action_type": "COARSE_ISOLATION",
      "target_asset": "DEV-OFFICE-LAPTOP-04",
      "blocked_flows": [],
      "preserved_flows": []
    }
  ],
  "clinical_impact": "LOW: All affected assets are non-clinical Tier 3. Rapid isolation
allowed.",
  "confidence": 0.95,
  "reason": "Case CASE-CORR-A6E68F contains 2 correlated finding(s). Max Risk Score:
0.72. Safety mode: NORMAL.",
  "approval_level": "AUTO_APPROVED",
  "expiry_time": "2026-10-07T14:03:48.233946+00:00",
  "rollback_plan": {
    "rollback_token_spec": "RB-CAMPAIGN-AF8192",
    "steps": [
      "Reconnect isolated endpoint 'DEV-OFFICE-LAPTOP-04'",
      "Verify telemetry heartbeat across all restored hosts"
    ],
```

```
    "auto_rollback_seconds": 300
  },
  "correlated_finding_ids": [
    "FIND-001",
    "FIND-002"
  ]
}
```


```
================================================================================
 >>> PROCESSING CAMPAIGN STAGE 3/6: Event Type 'CREDENTIAL_DUMP' in Zone 'radiology'
================================================================================

--- [1] LAYER 0: NORMALIZED OBSERVATION EVENT ---
```


```
{
  "event_id": "EVT-17083526",
  "timestamp": "2026-10-07T10:03:00+00:00",
  "source": "pacs_auth_sensor",
  "event_type": "CREDENTIAL_DUMP",
  "subject": "10.0.4.88",
  "src": "10.0.4.88",
  "dst": "10.0.4.10",
  "protocol": "TCP",
  "bytes": 3400,
  "zone": "radiology",
  "scenario_id": "six_stage_attack_campaign",
  "raw_reference": "/logs/raw/six_stage_attack_campaign/EVT-17083526.json"
}
```


```
[Layer 1 Error]: Asset IP/Name '10.0.4.88' not recognized in inventory.

================================================================================
 >>> PROCESSING CAMPAIGN STAGE 4/6: Event Type 'MALWARE_STAGING' in Zone 'laboratory'
================================================================================

--- [1] LAYER 0: NORMALIZED OBSERVATION EVENT ---
```


```
{
  "event_id": "EVT-23B5C6EC",
  "timestamp": "2026-10-07T10:05:00+00:00",
  "source": "lab_pcap_sensor",
  "event_type": "MALWARE_STAGING",
  "subject": "10.0.3.15",
  "src": "10.0.3.15",
  "dst": "10.0.2.10",
  "protocol": "TCP",
  "bytes": 8192,
  "zone": "laboratory",
  "scenario_id": "six_stage_attack_campaign",
  "raw_reference": "/logs/raw/six_stage_attack_campaign/EVT-23B5C6EC.json"
}
```


```
--- [2] LAYER 1: ASSET CONTEXT & CLINICAL TIER ---
```


```
{
  "asset_id": "DEV-OFFICE-LAPTOP-04",
  "asset_type": "Reception Desk Endpoint",
  "owner_zone": "Administration",
```

```
  "vendor": "Dell",
  "software_version": "Windows 11",
  "clinical_tier": 3,
  "known_constraints": [
    "Non-clinical endpoint"
  ],
  "allowed_peers": [
    "10.0.3.1"
  ],
  "normal_services": [
    "HTTPS",
    "SMB"
  ],
  "fallback_mode": "NONE",
  "identity_confidence": 0.99,
  "timestamp": "2026-10-07T10:05:00+00:00"
}

--- [3] LAYER 2: DETECTION FINDING ---
{
  "finding_id": "FIND-004",
  "asset_id": "DEV-OFFICE-LAPTOP-04",
  "attacker_src": "10.0.3.15",
  "behavior_window": {
    "start": "2026-10-07T10:00:00+00:00",
    "end": "2026-10-07T10:09:00+00:00"
  },
  "anomaly_score": 0.85,
  "risk_score": 0.85,
  "indicators": [
    "MALWARE_STAGING_PAYLOAD",
    "SUSPICIOUS_DB_QUERY"
  ],
  "baseline_version": "v1.0.0",
  "confidence": 0.95,
  "related_events": [
    "EVT-23B5C6EC"
  ],
  "recommended_observation_period": 60,
  "timestamp": "2026-10-07T10:05:00+00:00"
}

--- [4] LAYER 3: CORRELATED CAMPAIGN RESPONSE PROPOSAL ---
Case ID:                    CASE-CORR-A6E68F
Correlated Affected Assets:  ['DEV-OFFICE-LAPTOP-04']
Correlated Finding Count:   3
Approval Level Required:     AUTO_APPROVED
{
  "case_id": "CASE-CORR-A6E68F",
  "threat_summary": "LATERAL CAMPAIGN DETECTED: Correlated 3 anomalies across assets
[DEV-OFFICE-LAPTOP-04]. Attacker origin(s): [10.0.3.15]. Potential multi-stage attack
chain in progress.",
  "affected_assets": [
    "DEV-OFFICE-LAPTOP-04"
```

```
  ],
  "candidate_actions": [
    {
      "action_type": "COARSE_ISOLATION",
      "target_asset": "DEV-OFFICE-LAPTOP-04",
      "blocked_flows": [],
      "preserved_flows": []
    }
  ],
  "clinical_impact": "LOW: All affected assets are non-clinical Tier 3. Rapid isolation
allowed.",
  "confidence": 0.95,
  "reason": "Case CASE-CORR-A6E68F contains 3 correlated finding(s). Max Risk Score:
0.85. Safety mode: NORMAL.",
  "approval_level": "AUTO_APPROVED",
  "expiry_time": "2026-10-07T14:03:48.236262+00:00",
  "rollback_plan": {
    "rollback_token_spec": "RB-CAMPAIGN-EEBECC",
    "steps": [
      "Reconnect isolated endpoint 'DEV-OFFICE-LAPTOP-04'",
      "Verify telemetry heartbeat across all restored hosts"
    ],
    "auto_rollback_seconds": 300
  },
  "correlated_finding_ids": [
    "FIND-001",
    "FIND-002",
    "FIND-004"
  ]
}

================================================================================
 >>> PROCESSING CAMPAIGN STAGE 5/6: Event Type 'EXPLOIT_ATTEMPT' in Zone 'intensive_care'
================================================================================

--- [1] LAYER 0: NORMALIZED OBSERVATION EVENT ---
{
  "event_id": "EVT-536D36D0",
  "timestamp": "2026-10-07T10:07:00+00:00",
  "source": "icu_pcap_sensor",
  "event_type": "EXPLOIT_ATTEMPT",
  "subject": "lab-server",
  "src": "lab-server",
  "dst": "10.0.1.45",
  "protocol": "TCP",
  "bytes": 4096,
  "zone": "intensive_care",
  "scenario_id": "six_stage_attack_campaign",
  "raw_reference": "/logs/raw/six_stage_attack_campaign/EVT-536D36D0.json"
}

--- [2] LAYER 1: ASSET CONTEXT & CLINICAL TIER ---
{
  "asset_id": "DEV-ICU-MONITOR-01",
```

```
  "asset_type": "Patient Monitor Gateway",
  "owner_zone": "Intensive Care Unit",
  "vendor": "Philips",
  "software_version": "v4.2.1",
  "clinical_tier": 1,
  "known_constraints": [
    "Never isolate completely",
    "Preserve vital feeds"
  ],
  "allowed_peers": [
    "10.0.1.5",
    "10.0.1.6"
  ],
  "normal_services": [
    "HL7_Vitals"
  ],
  "fallback_mode": "LOCAL_SD_CARD",
  "identity_confidence": 0.98,
  "timestamp": "2026-10-07T10:07:00+00:00"
}

--- [3] LAYER 2: DETECTION FINDING ---
{
  "finding_id": "FIND-005",
  "asset_id": "DEV-ICU-MONITOR-01",
  "attacker_src": "lab-server",
  "behavior_window": {
    "start": "2026-10-07T10:00:00+00:00",
    "end": "2026-10-07T10:09:00+00:00"
  },
  "anomaly_score": 0.94,
  "risk_score": 0.94,
  "indicators": [
    "EXPLOIT_ATTEMPT_SMB",
    "CRITICAL_FLOW_INTERRUPTION_RISK"
  ],
  "baseline_version": "v1.0.0",
  "confidence": 0.95,
  "related_events": [
    "EVT-536D36D0"
  ],
  "recommended_observation_period": 60,
  "timestamp": "2026-10-07T10:07:00+00:00"
}

--- [4] LAYER 3: CORRELATED CAMPAIGN RESPONSE PROPOSAL ---
Case ID:                    CASE-CORR-7CEA8A
Correlated Affected Assets:  ['DEV-ICU-MONITOR-01']
Correlated Finding Count:   1
Approval Level Required:     REQUIRES_HUMAN_APPROVAL
{
  "case_id": "CASE-CORR-7CEA8A",
  "threat_summary": "Detected anomalous behavior (EXPLOIT_ATTEMPT_SMB,
CRITICAL_FLOW_INTERRUPTION_RISK) targeting asset DEV-ICU-MONITOR-01.",
```

```
  "affected_assets": [
    "DEV-ICU-MONITOR-01"
  ],
  "candidate_actions": [
    {
      "action_type": "SURGICAL_BLOCK",
      "target_asset": "DEV-ICU-MONITOR-01",
      "blocked_flows": [
        {
          "protocol": "TCP",
          "port": 445,
          "direction": "INBOUND"
        }
      ],
      "preserved_flows": [
        {
          "protocol": "HL7",
          "port": 2575,
          "direction": "BIDIRECTIONAL"
        }
      ]
    }
  ],
  "clinical_impact": "CRITICAL: Campaign involves Tier 1 Life-Critical asset(s). Surgical
containment required to preserve vital data feeds.",
  "confidence": 0.95,
  "reason": "Case CASE-CORR-7CEA8A contains 1 correlated finding(s). Max Risk Score:
0.94. Safety mode: NORMAL.",
  "approval_level": "REQUIRES_HUMAN_APPROVAL",
  "expiry_time": "2026-10-07T14:03:48.237576+00:00",
  "rollback_plan": {
    "rollback_token_spec": "RB-CAMPAIGN-A7E002",
    "steps": [
      "Restore firewall table for Tier 1 asset 'DEV-ICU-MONITOR-01'",
      "Verify telemetry heartbeat across all restored hosts"
    ],
    "auto_rollback_seconds": 300
  },
  "correlated_finding_ids": [
    "FIND-005"
  ]
}

================================================================================
 >>> PROCESSING CAMPAIGN STAGE 6/6: Event Type 'C2_BEACON' in Zone 'intensive_care'
================================================================================

--- [1] LAYER 0: NORMALIZED OBSERVATION EVENT ---
{
  "event_id": "EVT-4A9AD94E",
  "timestamp": "2026-10-07T10:09:00+00:00",
  "source": "icu_pcap_sensor",
  "event_type": "C2_BEACON",
  "subject": "icu-monitor-gw",
```

```
  "src": "icu-monitor-gw",
  "dst": "10.0.1.50",
  "protocol": "TCP",
  "bytes": 1024,
  "zone": "intensive_care",
  "scenario_id": "six_stage_attack_campaign",
  "raw_reference": "/logs/raw/six_stage_attack_campaign/EVT-4A9AD94E.json"
}
[Layer 1 Error]: Asset IP/Name 'icu-monitor-gw' not recognized in inventory.

================================================================================
🚨 [LAYER 5: DECISION GATE - HUMAN APPROVAL REQUIRED]
================================================================================
⚠  A high-risk action requires human approval for Tier 1 asset: 'DEV-ICU-MONITOR-01'
📌 Case ID:                    CASE-CORR-7CEA8A
🔍 Threat Summary:             Detected anomalous behavior (EXPLOIT_ATTEMPT_SMB,
CRITICAL_FLOW_INTERRUPTION_RISK) targeting asset DEV-ICU-MONITOR-01.
🎯 Correlated Affected Assets: DEV-ICU-MONITOR-01
🏥 Clinical Impact:            CRITICAL: Campaign involves Tier 1 Life-Critical asset(s).
Surgical containment required to preserve vital data feeds.
🛡️ Candidate Actions:
   - SURGICAL_BLOCK on 'DEV-ICU-MONITOR-01'
     Blocked Flows:   [{'protocol': 'TCP', 'port': 445, 'direction': 'INBOUND'}]
     Preserved Flows: [{'protocol': 'HL7', 'port': 2575, 'direction': 'BIDIRECTIONAL'}]
🔄 Rollback Token:             RB-CAMPAIGN-A7E002
================================================================================
```

Input: Human Validation

```
   Input ->
     👉 Do you approve this proposal? (y/n): n
👤 Enter SOC Analyst Reviewer Name/ID [default: analyst_dr_smith]: Soumadip
📝 Enter Reason for Decision [default: Rejected by SOC Analyst due to risk of clinical
workflow disruption.]: Patient's dylasis is going on
```

Final Output: Post-Validation Execution

```
   Final Output ->
     --- [5] LAYER 5: FINAL DECISION RECORD RECORDED ---
{
  "decision_id": "DEC-HUMAN-CAMPAIGN-01",
  "case_id": "CASE-CORR-7CEA8A",
  "reviewer": "Soumadip",
  "decision": "REJECTED",
  "scope_hash": "e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855",
  "reason": "Patient's dylasis is going on",
  "timestamp": "2026-10-07T10:10:00.000000+00:00",
  "expiry": "2026-10-07T14:03:48.237576+00:00",
  "asset_id": "DEV-ICU-MONITOR-01",
  "approved_action": null
}

================================================================================
--- [6] LAYER 4: ENFORCEMENT ENGINE EXECUTION ---
================================================================================
⚠ Enforcement Skipped due to REJECTED Human Approval.
{
  "action_id": "ACT-CAMPAIGN-EXEC-01",
  "case_id": "CASE-CORR-7CEA8A",
  "decision_id": "DEC-HUMAN-CAMPAIGN-01",
  "target_asset": "DEV-ICU-MONITOR-01",
  "status": "SKIPPED_REJECTED",
  "applied_controls": [],
  "blocked_flows": [],
  "preserved_flows": [],
  "start_time": "2026-10-07T10:10:05.000000+00:00",
  "expiry_time": "2026-10-07T14:03:48.237576+00:00",
  "verification": {
    "malicious_flow_blocked": false,
    "preserved_clinical_flows_available": true,
    "checked_at": "2026-10-07T10:10:10.000000+00:00"
  },
  "rollback_token": "RB-NONE"
}

================================================================================
--- [7] LAYER 6: COMPLIANCE, REDACTION & EVIDENCE EXPORT ---
================================================================================
📄 Incident Timeline Report written to: ['artifacts/full_run/RPT-CASE-CORR-7CEA8A.md',
'artifacts/full_run/RPT-CASE-CORR-7CEA8A.json']
📦 Evidence Package exported with ID: EVID-CASE-CORR-7CEA8A
🔒 SHA-256 Tamper-Evident Audit Chain verified and saved to disk.
(agentic-ai-cyberdefence) soumadipsen@pop-os:~/Desktop/AI Engineering/agentic-ai-
cyberdefence$
```